ANALYTICAL METHODS:

1 DESCRIPTION

1.1 Physically examine a representative sample of the material for Appearance and other characteristics: it is **A white or almost white, crystalline powder, freely but slowly soluble in water, very slightly soluble in ethanol.**

2 IDENTIFICATION

2.1 **SPECIFIC OPTICAL ROTATION**

PROCEDURE

Accurately weigh and transfer 10.0g of anhydrous sample into a 100ml volumetric flask.

2.1.1 Add 60ml of distilled water, 0.2ml of 6M ammonium hydroxide, mechanically shake/sonicate till dissolution is complete.
Dilute to the mark with distilled water, mix and allow standing for 30 minutes.
Measure the optical rotation of the blank solution prepared by diluting 0.2ml of 6M ammonium hydroxide to 100.0ml with distilled water, making at least 5 readings and determine the average of the 5 readings at 25°C.
Immediately observe the optical rotation of the prepared sample solution, making at least 5 readings and determine the average of the 5 readings at 25°C.
Subtract average of the blank reading (i.e. solvent readings) from that of the sample reading, the value thereby obtained is the corrected observed angle of rotation.

Calculation:

$$[\alpha]_D^{25} = \frac{100\,(\alpha 1 - \alpha 2)}{\text{Length Of Polarimeter tube (dm) X Sample Weight (g)}}$$

Where: $[\alpha]_D^{25}$ = specific rotation

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$\alpha 1$ = observed rotation of sample preparation at temperature t and wavelength
$\alpha 2$ = observed rotation of blank preparation at temperature t and wavelength

3 APPEARANCE OF SOLUTION

3.1 Dissolve 10.0 g of sample in 15 mL of *water R and mix*

3.2 The solution is clear and not more intensely coloured than reference solution.

4 CONDUCTIVITY

4.1 Maximum 20 µS/cm.

4.2 Accurately weigh and transfer about 20g of sample into a 100ml volumetric flask.

4.3 Add 40ml carbon dioxide-free distilled water and mechanically shake/sonicate till dissolution is complete.

4.4 Dilute to the mark with distilled water and mix.

4.5 Measure the conductivity of the solution while gently stirring with a magnetic stirrer

5 5.1 **DEXTRIN**

5.2 Accurately weigh and transfer about 1g of sample into a 100ml flask

5.3 Add 20ml of *ethanol* (96% *)*

5.4 heat under a reflux condenser.

5.5 The substance dissolves completely.

6 **SOLUBLE STARCH, SULFITE**

6.1 Maximum 15 ppm.

6.2 Accurately weigh and transfer about 7.4g of sample into a 100ml beaker.

6.3 Add 15ml of carbon dioxide free distilled water

6.4 heating on a water bath Allow to cool and

6.5 add 25 µL of iodine solution

6.6 The solution is yellow.

7 **WATER**

7.1 Weigh a labelled petri dish (W1) that has previously been dried at 80°C and cooled.

7.2 Accurately weigh about 1g onto a petri-dish, ensuring they are evenly spread, then determine their weight together (W2).

7.3 Transfer into an Oven operating at 80°C.

7.4 Dry for 2 hours

7.5 Transfer into a desiccators to cool room temperature.

7.6 Weigh the petri-dish with its content (W3

7.7 Calculate the moisture content as follows:

$$\% \text{ Moisture} = \frac{(W2 - W3) \times 100}{(W2 - W1)}$$

Where, W1 = weight of empty petri-dish, g.

W2 = weight of sample and petri-dish, g.

W3 = weight of petri-dish and the dried sample, g.

8 Microbiological Tests

8.1 <u>Limulus Amebocyte Lysate LAL Test</u>

8.1.1 Open the seals of the test tubes of lyophilized LAL Reagent

8.1.2 Add 0.2ml LAL Reagent water, endotoxin standards or test sample into each tubes.

8.1.3 Swirl gently but thoroughly to rehydrate the LAL reagent. Do not shake or vortex as content will foam.

NOTE: cover the reaction tubes with pyrogen free aluminium foil

8.1.4 Incubate the reaction mixture immediately at 37°c for 60minutes. Avoiding vibration during incubation.

8.1.5 If large numbers of samples are tested in parallel, the tests should be batched and started at intervals that permit the reading of each within the time limit.

8.1.6 Sample to be tested should be stored under conditions that bacteriological activities are controlled

8.1.7 Read reaction tubes one by one.

8.1.8 Invert the tube in one smooth motion.

8.1.9 A positive result is noticed by the formation of a gel which does not collapse when the tube is inverted. A negative result is characterized by

the absence of solid clot after inversion.

8.2

Testing Methodology of Bacterial Endotoxin (LAL)

8.2.1 Each specimen shall be tested in duplicate

8.2.2 NMC: Take 100µl(0.1ml) LRW

8.2.3 Take 50µl (0.05ml) LRW and 50µl (0.05ml) test dilution.

8.2.4 Take 50µl(0.05ml) test dilution and 50µl(0.05ml) CSE of 4λ

8.2.5 Add the LAL reagents as mentioned below in the table 1

8.2.6 Mix gently and incubate the tubes in heating block at 37^0C ± 1 for 60min.

8.2.7 After incubation, remove the tubes gently from the heating block and slowly invert through 180^0C and note the observations.

8.2.8 If a firm gel has formed, record the results as positive.

8.2.9 A result is negative, if an intact gel is not formed.

8.2.10 Record the results and other details as per format no. "Analytical Test Data Sheet for Bacterial Endotoxin (LAL) Test" in case of sterile raw material.

8.3

Pyrogen

8.3.1 Perform the test in a separate area designated for Pyrogen testing and under environmental conditions similar to those under which the rabbits are housed.

8.3.2 Withhold all food from the rabbits used during the period of the test

8.3.3 Draw out the required number of rabbits from their living room according to the number of tests planned. Always select more rabbits than actually required to counter any exclusion of animals in the initial measuring.

8.3.4 All the rabbits which are to be used for the test should be more than 1.5kg weight, should be healthy and must not have been used for any test in previous three(3) days and in previous three (3)weeks if any test failed using that rabbit.

8.3.5 Put the selected rabbits to testing cages. Take the initial temperature of each rabbit by inserting the thermometer to in the anus rectum of the rabbit to a depth of about 7.5cm.

8.3.6 If the initial temperature is less than 38.8^{0}c or more than 39.8^{o}cexclude the animal from the test.

8.3.7 Ensure all the needles and syringe to be used are new

8.3.8 Arrange the rabbits into groups of three taking into consideration that the initial temperature of the rabbits in a group should not differ by more than 1^{0}c.

8.3.9 Take another two reading of temperature in 30 min intervals between 1 and 3 hours subsequent to the injection.

8.3.10 Any rabbits showing more than 0.2^0c temperature difference between two successive readings in the last three readings can be excluded from the test.

8.3.11 Calculate the initial temperatures of the animals by taking the average of the temperature readings taken in the last 40 minutes.

8.3.12 Inject the solution intended to be tested to the marginal veins of the rabbits of a group slowly but not exceeding 10 minutes after the start of administration. The quantity of test solution depends upon the body weight of the rabbits and the test solution.

8.3.13 If the substance to be tested is not in aqueous sterilized solution form, consult the official monograph regarding the preparation of simulating solution and the volume of the preparation to be injected.

8.3.14 After injecting the solution, Record the temperature at 30 minutes intervals between 1 and 3 hours subsequent to the injection.

8.3.15 The test involves measuring the rise in temperature of rabbits following the intravenous injection of a test solution

8.4 MLT

8.4.1 Weigh 1g or 1ml of the raw material sample and dissolve aseptically in 9ml sterile soybean casein digest medium prepared as per SOP for media preparation (BML-QDG-MC-011).

8.4.2 Determine the Total Aerobic Microbial count (TAMC) and Total Yeast and Molds count (TYMC) by using pour plate method.

8.4.3 Testing to be performed under LAF of specified area

8.4.4 Arrange all the required material on the LAF before starting the test.

8.4.5 Aseptically Pipette 1ml of the solution from above in a sterile Petri dishes

8.4.6 Add approximately 15-20ml of sterile Soyabean Casein Digest Agar prepared as per SOP(BML-QDG-MC-011), (cooled approximately up to 45°C) in two petri plates for TAMC

8.4.7 Add approximately 15-20ml of sterile Saboraud Dextrose Agar prepared as per SOP (cooled approximately up to 45°C) in two petri- plates for TYMC

8.4.8 Gently swirl the contents inside the petri-dishes for proper mixing of the sample, allow medium to solidify, taking the precaution of not spilling the media from the plate

8.4.9 Simultaneously perform the negative control by plating 1 ml of diluents used instead of the sample for each medium. Add 15-20ml of the media

previously maintained not more than 45°c and allow for solidification.

8.4.10 Incubate the plates at 30°C to 35°C for 3 - 5 days for soybean Casein Digest Agar for TAMC and plates of Saboraud Dextrose Agar at 20^{0}C to 25°C for 5 days in an inverted position.

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